👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD MPP.7

Math is Playful

Lesson 10-3 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

Language Objective: I can describe a puzzle strategy using the words pattern, rule, and predict.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

puzzle · **pattern** · **pattern rule** · **predict** · **organize**

Level 3: try the blanks from memory first, then check the bank.

1. A problem designed to be played with — rules, a goal, room to imagine — is a _____.

2. Objects or shapes repeated in a regular order form a _____.

3 The rule for getting the next value from the one before is the _____ _____ .

4 To say what will happen before it happens, using a pattern, is to _____ .

5 To arrange information so patterns become easier to see is to _____ it.

Write About the Math

How can a pattern help you imagine the solution?
¿Cómo te puede ayudar un patrón a imaginar la solución?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **puzzle** — *rompecabezas*
- ☐ **pattern** — *patrón*
- ☐ **pattern rule** — *regla del patrón*
- ☐ **predict** — *predecir*
- ☐ **organize** — *organizar*

Model: *The pattern rule is to double the previous number of moves and add 1: 1 × 2 + 1 = 3, and 3 × 2 + 1 = 7.* — A strong answer states 'double and add 1' and checks it against both jumps (1→3 and 3→7). A weak answer only notices the numbers are growing without naming the rule.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The pattern rule for the Tower of Hanoi is to ___ the number of steps for the smaller tower and ___ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of ___ steps.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** puzzle
2. **Notes 2:** pattern
3. **Notes 3:** pattern rule
4. **Notes 4:** predict
5. **Notes 5:** organize
6. **Watch for this mistake:** *Doubling the previous number of steps but forgetting to add 1 — answering 14 instead of 15 for four discs, or 30 instead of 31 for five.*
7. **Try It 1:**

8. **Try It 2:**